TrustShift: Shift Type, Not Shift Magnitude, Determines Machine-Learning Failure Modes

A cross-domain audit protocol and a failure taxonomy

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Abstract

Deployment evaluation implicitly assumes that a larger distribution shift means greater risk. Across four domains, we find this assumption false. We introduce **TrustShift**, a single pre-registered protocol measuring discrimination, calibration, and subgroup reliability, and apply it identically to four real deployment shifts: clinical risk prediction (NHANES→BRFSS), mental-health text classification (Kaggle→Reddit/Twitter), mortgage-approval prediction (temporal and cross-state shift over 42M U.S. loan records), and network-intrusion detection (CIC-DDoS2019→CICIDS2017). A lending model under large measured covariate shift (domain-classifier AUC up to 0.80) degrades on no axis, while a mental-health model at comparable magnitudes loses up to 0.39 AUC: magnitude does not predict damage. The shift *type* does. Concept shift breaks every axis; novel-label shift blinds a detector to unseen classes while its in-domain AUC reads a perfect 1.00; covariate shift alone breaks nothing; mixed shift breaks calibration silently while discrimination holds. In-domain metrics anticipated none of this, yet three inexpensive diagnostic probes correctly distinguish the shift type before deployment. Post-hoc recalibration then cuts calibration error by one to two orders of magnitude in every domain but never restores lost discrimination or subgroup reliability. We contribute a protocol and a four-domain benchmark, and distill the result into a failure taxonomy mapping a pre-deployment diagnosis to the trustworthiness axis at risk.

1 Introduction

Consider three machine-learning systems at the moment of deployment. A clinical risk model moves from a survey cohort to a national screening population; its discrimination barely changes, and it looks fine—yet its probability estimates become substantially miscalibrated, so the risk numbers it reports are wrong. An intrusion detector moves from one year's attack corpus to the next; its in-domain AUC is a perfect 1.00, and it looks flawless—yet it is nearly blind to the attack families it will actually meet. A mortgage-approval model moves across years and states under a distribution shift large enough to be detected almost perfectly by a classifier trained to spot it—and nothing breaks; its accuracy even improves.

Three systems, three intuitions overturned. We argue these are not isolated accidents but three faces of one deployment phenomenon. The one that looked fine failed; the one that looked perfect failed; the one that looked most shifted did not. Why? Answering that question is the subject of this paper, and the answer is not the magnitude of the shift.

Deployment evaluation is dominated by a single in-domain number—accuracy, or AUC—treated as evidence of readiness. When shift is considered at all, it is treated as a scalar severity: bigger shift, bigger risk. And it is studied one domain at a time. Clinical prediction has a mature external-validation literature; fairness-under-shift has been examined in medical imaging [Schrouff et al., 2022]; tabular shift has dedicated benchmarks [Gardner et al., 2023]; general shift benchmarks span vision and language [Koh et al., 2021]. Each answers "does this model degrade here?" None answers the cross-domain question our three vignettes raise: *are deployment failures of trustworthiness domain-specific accidents, or do they follow rules that hold across domains—and can those rules be read before deployment?*

We answer it with one protocol applied to four deliberately dissimilar domains (the three above plus mental-health text classification). The protocol measures three trustworthiness axes (discrimination, calibration, subgroup reliability), assigns each shift a mechanistic diagnosis (label, covariate, or concept shift) using cheap and largely label-free probes, and quantifies what a remediation ladder restores. Holding the protocol fixed across clinical tabular data, social-media text, mortgage records, and network traffic separates what is general from what is idiosyncratic. The result is a taxonomy (Figure 1): across these domains the shift type, not its magnitude, consistently predicts which axis fails, and the shift type is cheap to diagnose in advance.

Contributions.

- **Shift type, not magnitude, tracks the failure (the discovery).** We show—and expose the statistical trap behind—that the *magnitude* of covariate shift does not predict deployment damage, while the shift *type* does (Section 4.3). A lending model at large measured shift is robust; a text model at comparable magnitude collapses.
- **A failure taxonomy backed by four domains.** We show that the trustworthiness axis that fails is consistently predicted by the shift's mechanistic type across four domains (Table 2)—turning measurement into classification.
- **A remediation result.** Across all four domains, post-hoc recalibration is the only cheap repair and it repairs only calibration; discrimination and subgroup reliability lost to shift require target supervision.
- **A protocol, not just a benchmark.** TrustShift is a reusable, pre-registered five-quantity audit (discrimination, calibration, subgroup gap, shift diagnosis, remediation) that runs identically on any (source, target) prediction pair, generalizing the single-domain CPFE protocol [Pall and Yadav, 2026] to a domain-agnostic instrument.
- **A released benchmark.** Four harmonized shift-pairs, standardized predictions, and the protocol implementation, enabling others to place new domains on the same axes.

2 The TrustShift Protocol

The protocol consumes, for each domain, a source model's predictions on an in-distribution test split (*source*) and on one or more deployment splits (*target*), together with a subgroup label per example. It emits five quantities.

Axis 1: Discrimination. AUC (macro one-vs-rest for multiclass domains) and macro-F1 per split, with DeLong 95% confidence intervals [DeLong et al., 1988]. We report $\Delta\text{AUC} = \text{AUC}_{\text{source}} - \text{AUC}_{\text{target}}$.

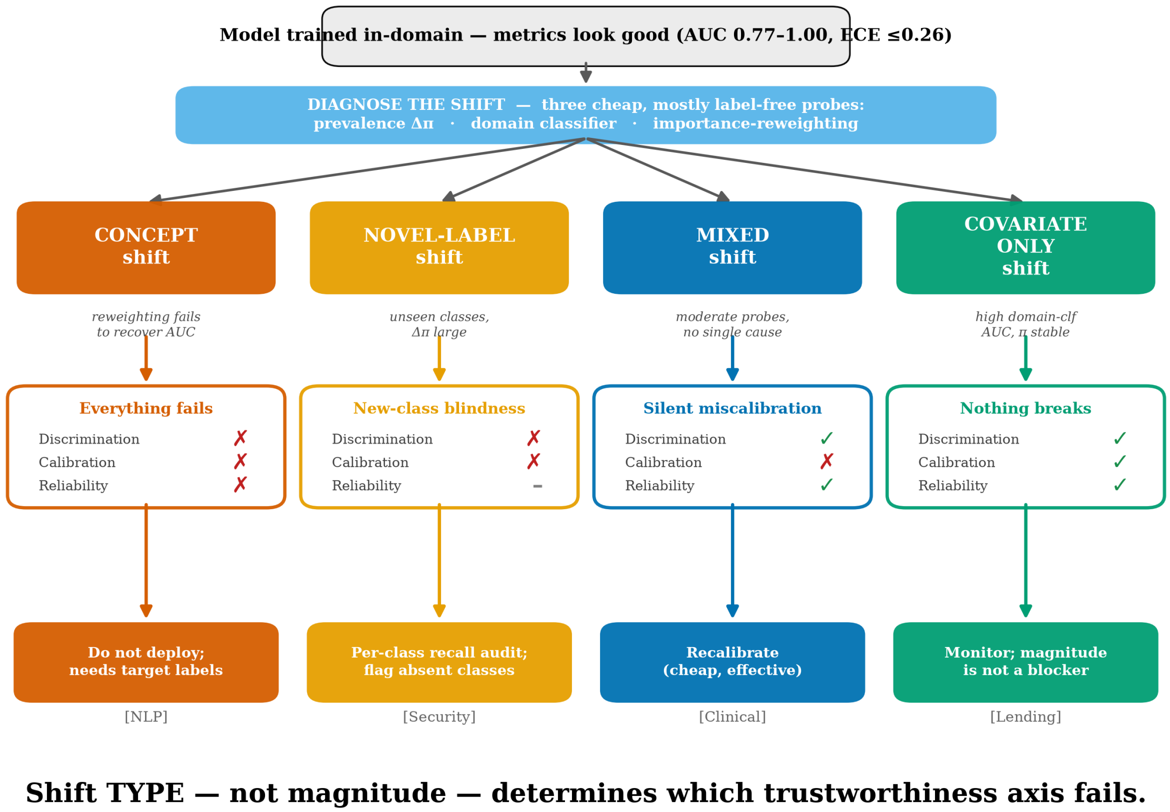

Figure 1: **The TrustShift failure taxonomy.** A model that looks deployable in-domain (top) is diagnosed by three cheap, largely label-free probes; the resulting shift type (colored branches) predicts which trustworthiness axis fails and the audit a practitioner should run. Each branch is realized by one of our four domains. In-domain metrics predicted none of these outcomes; the shift type did.

Axis 2: Calibration. Expected Calibration Error (ECE, 10 equal-width bins) and Brier score per split. ECE quantifies the gap between predicted confidence and empirical accuracy [Guo et al., 2017]; a model can discriminate perfectly and still be badly calibrated, and vice versa.

Axis 3: Subgroup reliability. Per subgroup g, AUC_g; the subgroup gap $G(\text{split}) = \max_g \mathrm{AUC}_g - \min_g \mathrm{AUC}_g$; and the *gap change* $\Delta G = G(\text{target}) - G(\text{source})$—the headline reliability quantity—with a stratified bootstrap 95% CI ($N = 2000$). Subgroups are demographic where available (clinical: age, sex, race, BMI; lending: race), platform-derived proxy classes for text, and *operational* (attack family) for intrusion detection. We call this subgroup-conditional reliability and, for intrusion detection, never interpret it as demographic fairness (Section 7).

Axis 4: Shift diagnosis. Each shift is decomposed with three cheap probes. *Label shift*: change in class prevalence $\Delta\pi$ (labels only). *Covariate shift*: a domain classifier trained to separate source from target examples on shared features, scored by 5-fold cross-validated AUC ($\mathrm{AUC}_{\mathrm{dc}}$; $0.5 =$ no shift, $\to 1 =$ severe) [Rabanser et al., 2019]. *Concept shift*: importance-weighting the source examples by the domain-classifier odds and recomputing source AUC; if the reweighted source AUC

Algorithm 1 The TrustShift audit

Require: source predictions $(\hat{p}, y, g)_{\text{src}}$; target predictions $(\hat{p}, y, g)_{\text{tgt}}$; shared features $X_{\text{src}}, X_{\text{tgt}}$

Stage 1 — Evaluate the three trustworthiness axes

$\Delta\text{AUC} \leftarrow \text{AUC}_{\text{src}} - \text{AUC}_{\text{tgt}}$; ECE, Brier per split ▷ DeLong CIs; 10 bins

$\Delta G \leftarrow (\max_g - \min_g)\text{AUC}_g^{\text{tgt}} - (\max_g - \min_g)\text{AUC}_g^{\text{src}}$ ▷ subgroup gap; bootstrap CI

Stage 2 — Diagnose the shift type

$\Delta\pi \leftarrow$ prevalence change; $\text{AUC}_{\text{dc}} \leftarrow$ CV-AUC of a source-vs-target classifier on X

$w(x) \leftarrow$ domain-classifier odds; $\text{AUC}_{\text{src}}^{\text{rw}} \leftarrow w$-weighted source AUC

if $\text{AUC}_{\text{src}}^{\text{rw}} \approx \text{AUC}_{\text{src}} \gg \text{AUC}_{\text{tgt}}$ **then** diagnosis $\leftarrow$ **concept**

else if $|\Delta\pi|$ large / unseen classes **then** diagnosis $\leftarrow$ **novel-label**

else if $\text{AUC}_{\text{dc}} > 0.65$ **then** diagnosis $\leftarrow$ **covariate**

else diagnosis $\leftarrow$ **mixed**

end if

Stage 3 — Remediate and re-measure

fit recalibration on 10% target labels; re-measure the three axes on the rest

Ensure: (ΔAUC, ECE, ΔG, diagnosis, restored axes)

Table 1: The four deployment shifts. Subgroup semantics deliberately differ; the protocol is identical.

Domain	Source → Target	Modality	Subgroup axis	Shift mechanism
Clinical	NHANES → BRFSS	tabular	age/sex/race/BMI	population
NLP	Kaggle → Reddit, Twitter	text	proxy class	platform/style
Lending	HMDA 2020–21 → 2022–24; cross-state	tabular	race, income	temporal, geographic
Security	CIC-DDoS2019 → CICIDS2017	network flow	attack family	novel attacks

stays near the source AUC while the target AUC is far lower, covariate shift alone cannot explain the drop [Lipton et al., 2018].

Axis 5: Remediation ladder. On a 10% stratified target calibration split we apply temperature, Platt, and isotonic recalibration [Zadrozny and Elkan, 2002], then re-measure Axes 1–3 on the remaining 90%. This isolates what a cheap post-hoc fix restores from what requires target supervision.

Full metric definitions, seeds, and estimator choices are fixed in the released configuration; all reported numbers regenerate from committed prediction files. Algorithm 1 states the protocol as executed.

3 Domains and Data

We chose four domains that differ in modality, subgroup semantics, and shift mechanism, so that any shared pattern is unlikely to be an artifact of one setting (Table 1).

The clinical and NLP source models reuse published predictions (Pall and Yadav, 2026, a preprint under review, for NLP, evaluated on GoEmotions [Demszky et al., 2020] and a Twitter emotion corpus; a diabetes external-validation study on NHANES [National Center for Health Statistics, Centers for Disease Control and Prevention, 2020] and BRFSS [Centers for Disease Control and Prevention, 2022] for clinical), so no retraining occurs and the source numbers reproduce

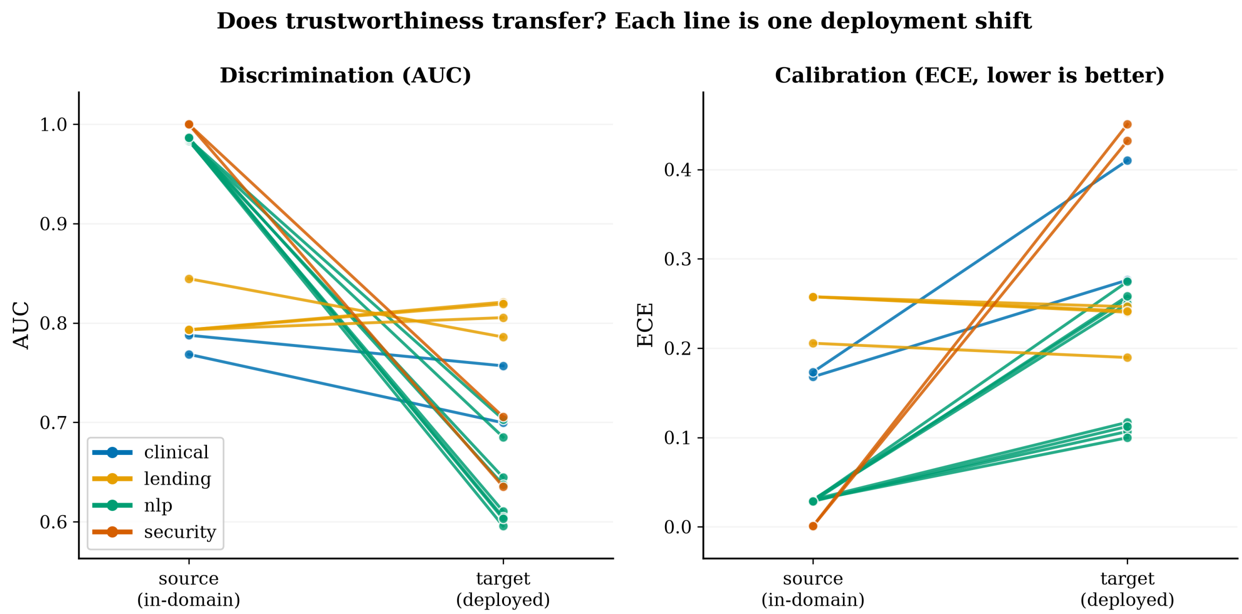

Figure 2: Discrimination (left) and calibration (right) as each shift transfers: every line runs from a shift-point's source (in-domain) value to its target (deployed) value, colored by domain. Source values are high and uniform; target values fan out. Steep lines mark collapse (NLP, security AUC), flat lines robustness (lending). The clinical lines are near-flat for AUC yet rise steeply for ECE—calibration fails where discrimination does not.

prior work. The lending and security analyses are new: for lending we train gradient-boosted approval classifiers with no race or geographic feature on HMDA loan records [Consumer Financial Protection Bureau, 2024] and evaluate temporal (train 2020–21, test each of 2022/2023/2024) and geographic (train on 60% of states, test on held-out states) shift over 42.3M records; for security we train intrusion detectors on CIC-DDoS2019 [Sharafaldin et al., 2019] and evaluate on CICIDS2017 [Sharafaldin et al., 2018]. Data are public. Clinical target-side subgroup analysis is limited to age for a reason we state plainly in Section 7.

4 Results

We organize results as one argument: in-domain metrics do not predict deployment behaviour (4.1); the failure that occurs is consistently predicted by the shift type across domains (4.2); that regularity is categorical, not a magnitude law (4.3); and recalibration repairs only calibration (4.4).

4.1 In-domain performance does not transfer, and does not transfer uniformly

Every source model looked deployable: source AUC ranged from 0.77 (clinical) to 1.00 (security) and source ECE was at most 0.26. Deployment outcomes did not follow (Figure 2, Table 3). Mental-health classifiers lost 0.28–0.39 AUC on new platforms; the intrusion detector fell from AUC 1.00 to 0.64; the clinical model barely moved in AUC (0.788 $\rightarrow$ 0.757); and lending models did not degrade at all, in fact improving slightly on later years. A single in-domain dashboard would have rated all four identically "ready." None of the deployment outcomes is readable from it.

Table 2: **The failure fingerprint.** One row per domain (primary model). ↑/↓ mark the direction and ≈ near-invariance. The trustworthiness axis that fails tracks the diagnosis, across four dissimilar domains.

Domain	Diagnosis	Discrimination	Calibration	Subgroup reliability
NLP	concept	collapses (↓ 0.28–0.39)	breaks	gap widens (↑)
Security	novel-label	collapses (1.00→0.64)	breaks (↑ 450×)	novel-family blind
Clinical	mixed	**holds** (≈)	**breaks silently** (↑)	gap narrows (↓)
Lending	covariate	holds/improves (≈)	stable (≈)	gap narrows (↓)

Table 3: Master results, primary models. Source and target AUC/ECE, discrimination change, and subgroup-gap change ΔG (positive = gap widens under shift). Numbers regenerate from committed prediction files.

Domain	Target	AUC_{src}	AUC_{tgt}	ΔAUC	ECE_{src}	ECE_{tgt}	ΔG
Clinical	BRFSS	0.788	0.757	+0.031	0.168	**0.276**	−0.066
NLP	Reddit	0.984	0.645	+**0.339**	0.030	0.117	+0.074
NLP	Twitter	0.984	0.596	+**0.388**	0.030	0.255	+0.080
Lending	2023	0.793	0.821	−0.028	0.258	0.240	−0.031
Lending	held-out states	0.845	0.786	+0.059	0.206	0.190	−0.029
Security	CICIDS2017	**1.000**	**0.635**	+**0.365**	0.001	**0.451**	—

4.2 The failure is consistently predicted by the shift type (the taxonomy)

The four domains produce four distinct and interpretable failure fingerprints (Table 2), and each fingerprint matches its mechanistic diagnosis.

Concept shift → everything fails. The NLP shift diagnoses as concept shift: the domain classifier separates platforms almost perfectly ($AUC_{dc} = 0.93$–0.99), yet importance-reweighting the source barely changes source AUC ($0.988 \to 0.987$) while target AUC is ≈ 0.60—covariate shift cannot explain the drop. Discrimination collapses (ΔAUC 0.28–0.39), calibration breaks (target ECE up to 0.27), and the subgroup gap widens (ΔG up to +0.15). All three axes fail together.

Novel-label shift → discrimination and calibration collapse on new classes. The intrusion detector achieves a perfect in-domain AUC of 1.00 (a known separability property of whole-flow DDoS features), then falls to 0.64 externally, with ECE rising from 0.001 to 0.45. The mechanism is label shift ($\Delta\pi = 0.34$): the deployment set contains attack families absent from training. Per-family recall exposes it directly—families seen in training are detected at 0.99+, while novel families are nearly invisible (`hulk` 0.05, `slowhttp` 0.12, `slowloris` 0.21). A detector can be perfect in-domain and blind to the attacks it will actually meet.

Covariate shift alone → no damage. The lending temporal shift diagnoses as covariate (AUC_{dc} 0.69–0.80) with no label or concept component, and nothing fails: AUC *improves* (ΔAUC −0.012 to −0.028), calibration is stable, and the race gap shrinks. Geographic shift behaves the same (mild ΔAUC = +0.059, gap −0.029).

Mixed → silent calibration failure. The clinical shift holds discrimination (ΔAUC = +0.031) while calibration deteriorates sharply (ECE $0.168 \to 0.276$ for the primary model; 0.41 for the boosted baseline). The age subgroup gap does not widen; it narrows ($\Delta G = -0.066$, 95% CI $[-0.113, -0.016]$). A model whose AUC certifies it as unchanged is producing risk estimates that are substantially miscalibrated.

4.3 The regularity is categorical, not a magnitude law

Treating each (domain, model, target) as a shift-point ($n = 16$), we asked whether the subgroup-reliability change is predictable from cheaper quantities. *Because the observations cluster strongly by domain (effectively four clusters), we interpret every regression in this section as an exploratory cross-domain summary, not a fitted predictor.* With that caveat stated first: across shift-points, gap change is associated with calibration change (slope 0.42, 95% CI $[0.17, 0.60]$, $R^2 = 0.41$) and with aggregate accuracy loss (slope 0.34, CI $[0.27, 0.44]$, $R^2 = 0.82$). The tempting third association—subgroup loss with covariate-shift magnitude—has a high cross-domain fit (slope 0.54, $R^2 = 0.77$) but is a statistical trap, which we report against ourselves.

Unexpected finding: shift magnitude is not deployment damage

The cross-domain regression of subgroup-gap change on covariate-shift magnitude ($\mathrm{AUC}_{\mathrm{dc}}$) has slope $+0.54$ with $R^2 = 0.77$—a fit that, taken at face value, would license "measure the shift; the bigger it is, the more subgroup reliability you lose." That reading is wrong. Within the lending domain, where covariate-shift magnitude rises from 0.69 to 0.80 across deployment years, the same slope is *negative* (-0.10): more measured shift, less damage. The positive cross-domain slope is a two-cluster (Simpson) artifact separating high-shift text from moderate-shift tabular data—an aggregate line drawn through two groups that individually tell the opposite story. Lending exhibits larger measured covariate shift than many settings described as "severe," while remaining fully robust on every axis. This mattered to our own analysis: an earlier version of this work reported the 0.54 slope as a finding before the within-domain check exposed it, and we retain it here precisely because the mistake is instructive. **Shift magnitude and deployment risk are not interchangeable**; the *categorical* diagnosis (Figure 1), recoverable from cheap probes, is what tracks failure.

We therefore report the cross-domain regressions (Figure 3) as exploratory synthesis over four domain-clusters, not as a fitted predictor, and rest the contribution on the categorical taxonomy.

4.4 Recalibration repairs calibration only

Isotonic recalibration on a 10% labeled target split reduced ECE by one to two orders of magnitude in every domain: clinical $0.276 \to 0.002$, lending $0.246 \to 0.006$, security $0.450 \to 0.013$, NLP $0.115/0.254 \to 0.016/0.055$ (Table 4). In the three continuous-score domains, AUC and the subgroup gap moved by less than 0.012—calibration is cheaply repairable, discrimination and reliability are not. (For security the isotonic map perturbs AUC by $+0.07$; because the in-domain-perfect detector emits near-degenerate probabilities, isotonic collapses them into 28 tied levels, and AUC with heavy ties is not transform-invariant. This is a tie artifact, not restored discrimination: $+0.07$ does not recover the 0.37 collapse.) Restoring discrimination requires target supervision—e.g., in the NLP domain, target fine-tuning recovers far more than any post-hoc fix [Pall and Yadav, 2026].

5 Discussion

What failed, and why. The four domains failed in four ways, and each way matched its shift mechanism. Concept shift (NLP) broke every axis because the input–label relationship itself changed. Novel-label shift (security) broke discrimination and calibration precisely on classes absent from training, leaving trained classes untouched. Covariate shift alone (lending) broke noth-

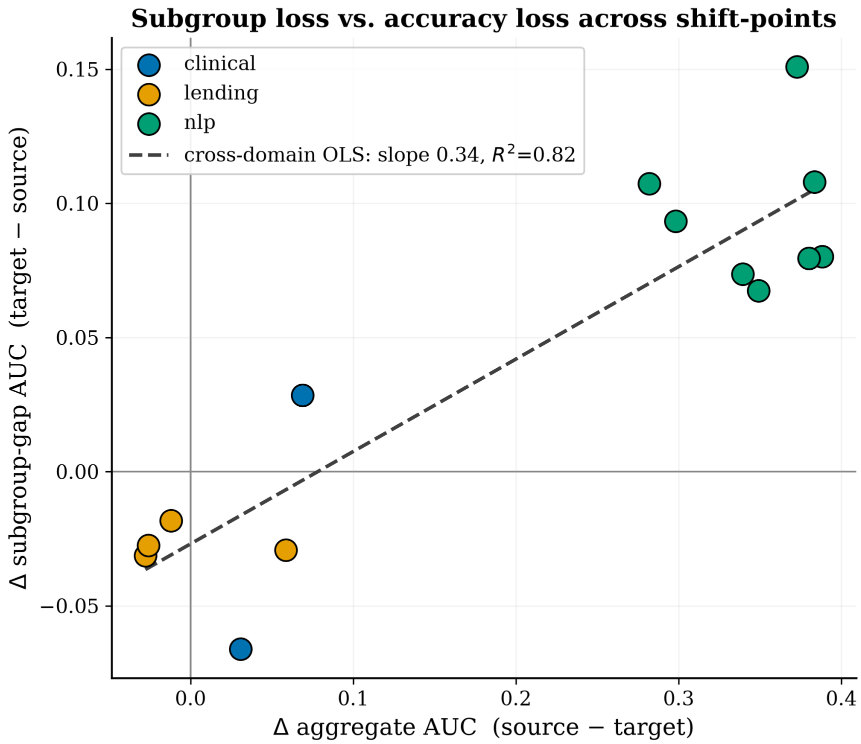

Figure 3: Subgroup-gap change vs. aggregate accuracy loss across shift-points, colored by domain. The cross-domain trend is real but domain-clustered; within-domain slopes differ (see box). Security is absent because its operational subgroups are all-positive, so a gap is undefined.

ing, even at large magnitude. Mixed shift (clinical) broke calibration while sparing discrimination. The common thread is mechanistic: the axis that fails is the axis the shift mechanism attacks.

The most practically important observation. Maintaining high discrimination after deployment did not imply trustworthy behaviour. The clinical model retained its AUC while its probability estimates became substantially miscalibrated, and the lending model stayed robust despite measurable covariate shift. These contrasting outcomes argue that deployment decisions should not rest on aggregate discrimination alone, but should incorporate diagnostics tailored to the underlying shift mechanism—which are cheap and, for covariate and label probes, require no target labels.

What practitioners should do differently. The results support a concrete pre-deployment routine: (1) probe label prevalence; (2) fit a source-vs-target domain classifier; (3) run the reweighting probe. The resulting diagnosis indicates which dashboard will mislead—a covariate-only diagnosis is reassuring, a concept or novel-label diagnosis is a warning that discrimination and subgroup reliability, not just calibration, are at risk—and a small labeled target sample should be budgeted for recalibration, which was cheap and universally effective. Monitoring aggregate AUC alone, the most common practice, was the least informative signal in all four domains.

Outlook. If this pattern continues to hold across additional domains, deployment evaluation may need to shift its emphasis from measuring *how large* a distribution shift is toward identifying *what type* of shift has occurred before deployment. We do not claim a universal law from four domains;

Table 4: Remediation ladder (isotonic recalibration on 10% target labels). ECE falls sharply; AUC is essentially unchanged in continuous-score domains.

Domain	Target	ECE (no fix)	ECE (isotonic)	AUC (no fix)	AUC (isotonic)
Clinical	BRFSS	0.276	0.002	0.757	0.757
Lending	2023	0.240	0.002	0.821	0.821
NLP	Twitter	0.254	0.055	0.592	0.589
Security	CICIDS2017	0.450	0.013	0.634	0.708†

† tie artifact from near-degenerate scores; not restored discrimination.

we claim a reproducible regularity, a mechanism that explains it, and a cheap protocol for testing whether it holds in a new setting. Establishing its boundaries—the domains, shift types, and model classes where the taxonomy breaks down—is the natural next step, and the benchmark is built to accept new domains.

6 Related Work

Shift benchmarks such as WILDS [Koh et al., 2021] and TableShift [Gardner et al., 2023] quantify how much accuracy drops under shift, within a modality. TrustShift differs in measuring calibration and subgroup reliability alongside discrimination, in assigning a mechanistic diagnosis, and in spanning modalities so cross-domain regularities are visible.

Fairness under shift has been examined mostly in single domains; Schrouff et al. [2022] show fairness properties fail to transfer in medical settings and that clinically plausible shifts violate the assumptions of single-mechanism mitigations. We corroborate the multi-mechanism point across four domains and connect the mechanism to which axis fails. A recent survey [Lin et al., 2024] catalogues methods; we contribute an empirical taxonomy.

Shift diagnosis and calibration. Our probes build on black-box shift detection [Rabanser et al., 2019], label-shift estimation [Lipton et al., 2018], and post-hoc calibration [Guo et al., 2017, Zadrozny and Elkan, 2002]. Our contribution is not a new estimator but the finding that these cheap diagnostics, applied jointly, predict *which* trustworthiness axis a deployment will break.

7 Limitations

Four domains and sixteen shift-points are evidence, not proof; the cross-domain regressions are exploratory synthesis over effectively four clusters, and we do not present them as a fitted predictor. Subgroup semantics differ across domains by necessity—demographic for clinical and lending, platform-derived proxy classes for text, and *operational* attack families for security—and we never interpret security's subgroup reliability as demographic fairness. Clinical target-side subgroup analysis is limited to age: the released prediction arrays were generated from a previously harmonized BRFSS dataset whose preprocessing pipeline is no longer recoverable, so sex, race, and BMI analyses are not reproducible on the target side; age-based analyses remain fully reproducible and constitute the primary clinical subgroup evaluation. The clinical and lending subgroup gaps *narrow* under shift, which we attribute tentatively to changes in subgroup composition rather than to genuine fairness improvement, and flag as a hypothesis. The security detector's near-perfect in-domain AUC is a known separability property of whole-flow features, which we treat as the in-domain ceiling rather than a result. This study examines four deployment settings rather than an exhaustive sample of machine-learning domains; important modalities—especially vision foundation models and large language models, whose shift behaviour may differ—remain to be tested, and the benchmark is designed to accept them. Finally, all experiments run on a single workstation, bounding model scale.

8 Conclusion

Deployment shift does not degrade machine-learning trustworthiness uniformly. Which axis fails—discrimination, calibration, or subgroup reliability—is set by the type of shift, and the type is recoverable before deployment with cheap, largely label-free diagnostics; the magnitude of shift is not, on its own, a measure of risk. In-domain metrics predicted none of this. We make the TrustShift protocol and benchmark available upon publication so that new domains can be placed on the same axes, and offer the resulting taxonomy as a practical guide to what a deployment audit should check.

Statements and Declarations

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Competing interests. The author declares no competing interests.

Ethics approval. Not applicable. This study uses only publicly available, de-identified datasets and involves no new human- or animal-subjects data collection.

Author contributions. Rajveer Singh Pall is the sole author and conducted all aspects of the work: conception, methodology, experiments, analysis, and writing.

Data availability statement. All source datasets are public: NHANES and BRFSS (U.S. Centers for Disease Control and Prevention), HMDA (U.S. Consumer Financial Protection Bureau), GoEmotions, CIC-DDoS2019, and CICIDS2017. Consistent with their licenses, the released benchmark redistributes standardized model *predictions* and metadata—not raw third-party records. The harmonized benchmark of predictions is available on the Hugging Face Hub at `https://huggingface.co/datasets/Rajveer-code/trustshift`.

Code availability. The TrustShift protocol, the four domain adapters, the audit engine, and every results file that backs a number in this paper—together with a single script that regenerates all tables and figures from the committed prediction files—are available as an open-source repository at `https://github.com/Rajveer-code/trustshift`. Seeds, estimator settings, and metric definitions are fixed in the released configuration.

References

Centers for Disease Control and Prevention. Behavioral risk factor surveillance system (BRFSS). `https://www.cdc.gov/brfss/`, 2022.

Consumer Financial Protection Bureau. Home mortgage disclosure act (HMDA) loan/application records. `https://ffiec.cfpb.gov/data-browser/`, 2024.

Elizabeth R. DeLong, David M. DeLong, and Daniel L. Clarke-Pearson. Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. *Biometrics*, 44(3):837–845, 1988.

Dorottya Demszky, Dana Movshovitz-Attias, Jeongwoo Ko, Alan Cowen, Gaurav Nemade, and Sujith Ravi. GoEmotions: A dataset of fine-grained emotions. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4040–4054, 2020.

Josh Gardner, Zoran Popovic, and Ludwig Schmidt. Benchmarking distribution shift in tabular data with TableShift. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, volume 70 of *PMLR*, pages 1321–1330, 2017.

Pang Wei Koh, Shiori Sagawa, Henrik Marklund, Sang Michael Xie, Marvin Zhang, Akshay Balsubramani, Weihua Hu, Michihiro Yasunaga, Richard Lanas Phillips, Irena Gao, Tony Lee, Etienne David, Ian Stavness, Wei Guo, Berton Earnshaw, Imran Haque, Sara Beery, Jure Leskovec, Anshul Kundaje, Emma Pierson, Sergey Levine, Chelsea Finn, and Percy Liang. WILDS: A benchmark of in-the-wild distribution shifts. In *Proceedings of the 38th International Conference on Machine Learning (ICML)*, volume 139 of *PMLR*, pages 5637–5664, 2021.

Yujie Lin, Dong Li, Chen Zhao, Xintao Wu, Qin Tian, and Minglai Shao. Supervised algorithmic fairness in distribution shifts: A survey. *arXiv preprint arXiv:2402.01327*, 2024.

Zachary C. Lipton, Yu-Xiang Wang, and Alexander J. Smola. Detecting and correcting for label shift with black box predictors. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, volume 80 of *PMLR*, pages 3122–3130, 2018.

National Center for Health Statistics, Centers for Disease Control and Prevention. National health and nutrition examination survey (NHANES). `https://www.cdc.gov/nchs/nhanes/`, 2020.

Rajveer Singh Pall and Sameer Yadav. Cross-platform generalisation failure in mental health natural language processing: A five-axis fairness audit of transformer models on social media, 2026. Preprint; author's prior work introducing the CPFE protocol.

Stephan Rabanser, Stephan Günnemann, and Zachary C. Lipton. Failing loudly: An empirical study of methods for detecting dataset shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 32, 2019.

Jessica Schrouff, Natalie Harris, Oluwasanmi Koyejo, Ibrahim Alabdulmohsin, Eva Schnider, Krista Opsahl-Ong, Alexander Brown, Subhrajit Roy, Diana Mincu, Christina Chen, Awa Dieng, Yuan Liu, Vivek Natarajan, Alan Karthikesalingam, Katherine Heller, Silvia Chiappa, and Alexander D'Amour. Diagnosing failures of fairness transfer across distribution shift in real-world medical settings. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, 2022.

Iman Sharafaldin, Arash Habibi Lashkari, and Ali A. Ghorbani. Toward generating a new intrusion detection dataset and intrusion traffic characterization. In *Proceedings of the 4th International Conference on Information Systems Security and Privacy (ICISSP)*, pages 108–116, 2018.

Iman Sharafaldin, Arash Habibi Lashkari, Saqib Hakak, and Ali A. Ghorbani. Developing realistic distributed denial of service (DDoS) attack dataset and taxonomy. In *2019 International Carnahan Conference on Security Technology (ICCST)*, pages 1–8, 2019. doi: 10.1109/CCST.2019.8888419.

Bianca Zadrozny and Charles Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pages 694–699, 2002. doi: 10.1145/775047.775151.